

CSCI 5840: Pre-Class Checklist

In-Class Challenge · Complete before class

Do this before class. You will not have time to fix it during the challenge. Your lab must be fully built and working from Labs 1–5. Then confirm each item.

1. Connectivity

- ☐ Every host (H1–H4) can ping the web server. H4 is IPv6-only, so test it over IPv6.
- ☐ All four hosts can reach each other and the web server after a fresh start of your lab.

2. Continuous Ping That Shows Failures

- ☐ On every host, you can run a continuous ping to the web server and leave it running in a terminal window.
- ☐ It visibly shows a failed packet. Test it: briefly disable a host interface (or a link), and confirm you can see the failure in the window (a “timeout” or “no answer” line, or a jump in the sequence numbers). Then re-enable it and confirm the ping recovers.
- ☐ You can keep four ping windows open and readable at once, alongside your work.
- ☐ Your terminal scrollback is large enough that you can scroll back through several minutes of ping output.

Ping options differ by operating system. Find the form that works on your host OS and try it now, for example:

Host OS	Continuous ping that shows failures
Linux	<code>ping -O -D -i 1 <address></code> (use <code>ping -6</code> for IPv6)
Windows	<code>ping -t <address></code> (use <code>ping -6 -t</code> for IPv6)
macOS	<code>ping <address></code> (use <code>ping6</code> for IPv6); failures show as gaps in <code>icmp_seq</code>

3. Your Tools

- ☐ NMAS is up and receiving telemetry from your routers.
- ☐ NSOT is reachable and current.
- ☐ Your automation framework can reach every router programmatically (no CLI).
- ☐ You can see live interface counters for devices in the network.

4. Know Your Own Network

- ☐ Which routing protocols you run on the access side and the core side, for IPv4 and IPv6.
- ☐ Where DHCP runs in your design.
- ☐ How your hosts get their default gateway.

Bring a laptop with everything already open.